

FORENSICS

1 - Foundations

STT



!! nos4a2
@sluzyp

AFSC19.EXE
@afsc19

ROAD MAP

1. Foundations
2. Disks, registries
3. Memory
4. ???

What is Forensics?



Static & Dynamic Analysis



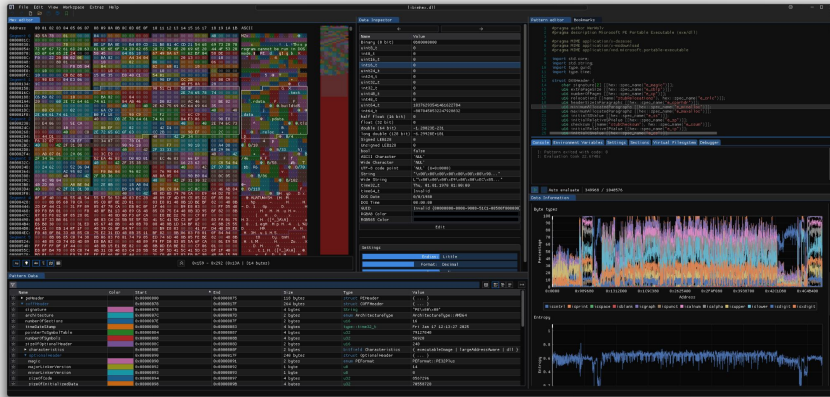
Static

VS

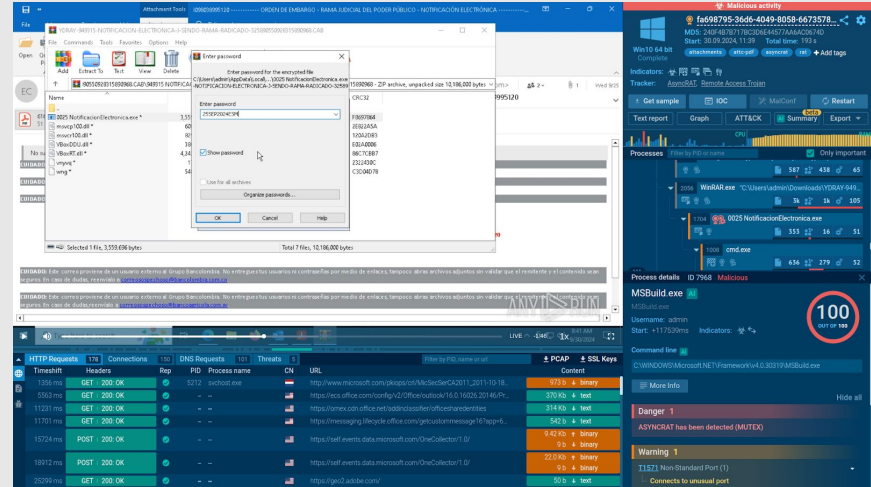


Dynamic

Static & Dynamic Analysis



VS



Static

Dynamic

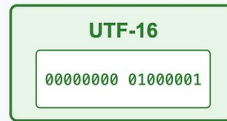
Data encoding schemes

UTF Encoding Comparison

Example: ASCII Character 'A' (U+0041)



1 byte

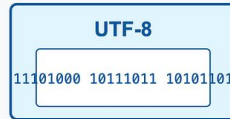


2 bytes

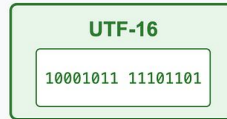


4 bytes

Example: Chinese Character '语' (U+8BED)



3 bytes

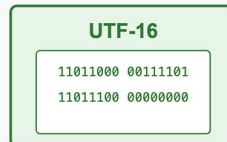
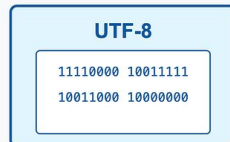


2 bytes



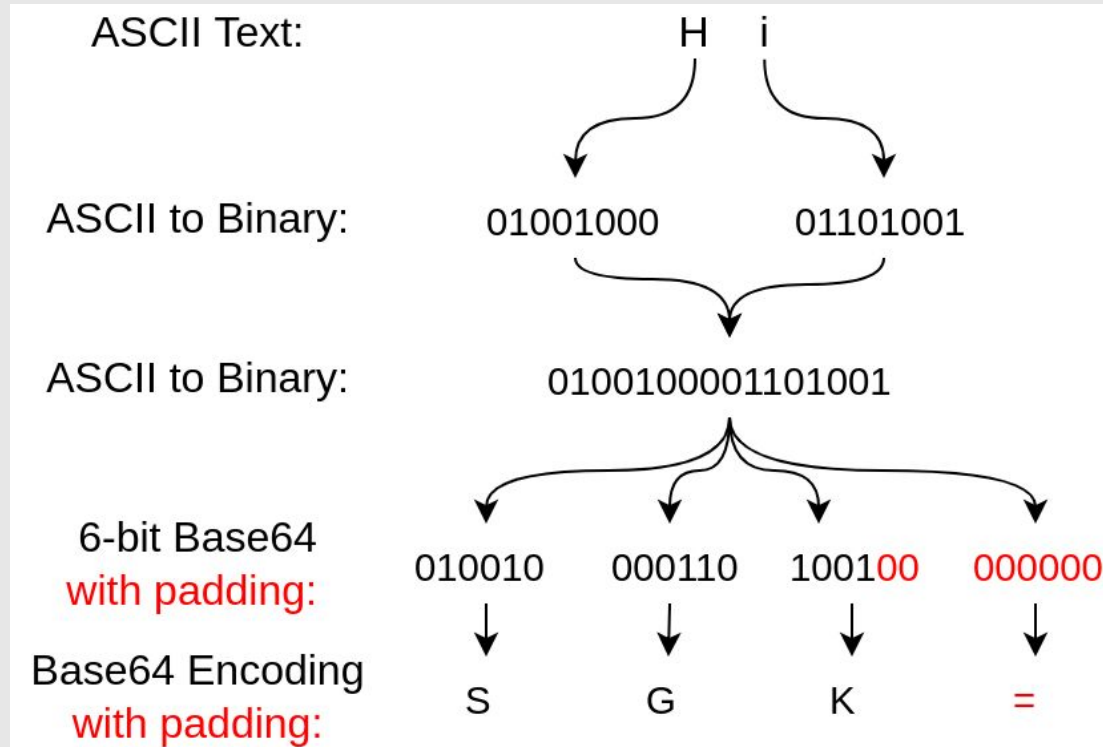
4 bytes

Example: Emoji '😄' (U+1F600)



Key Insight: UTF-8 is efficient for Western text, UTF-16 can be better for Asian languages, while UTF-32 offers consistent sizing at the cost of more storage space

Data encoding schemes



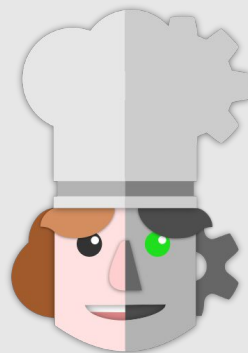
Data encoding schemes

Infinite encodings!

```
echo -n "1234567" | base16384 -e - - | iconv -f utf-16be -t utf-8  
𪛗𪛘𪛙𪛚𪛛𪛜𪛝
```


BUT WHAT ENCODING IS THIS?

CyberChef it!



PS: Check entropy!

High Entropy = Compressed data

CYBERCHEF ISN'T ALL!



, ...

What are files?

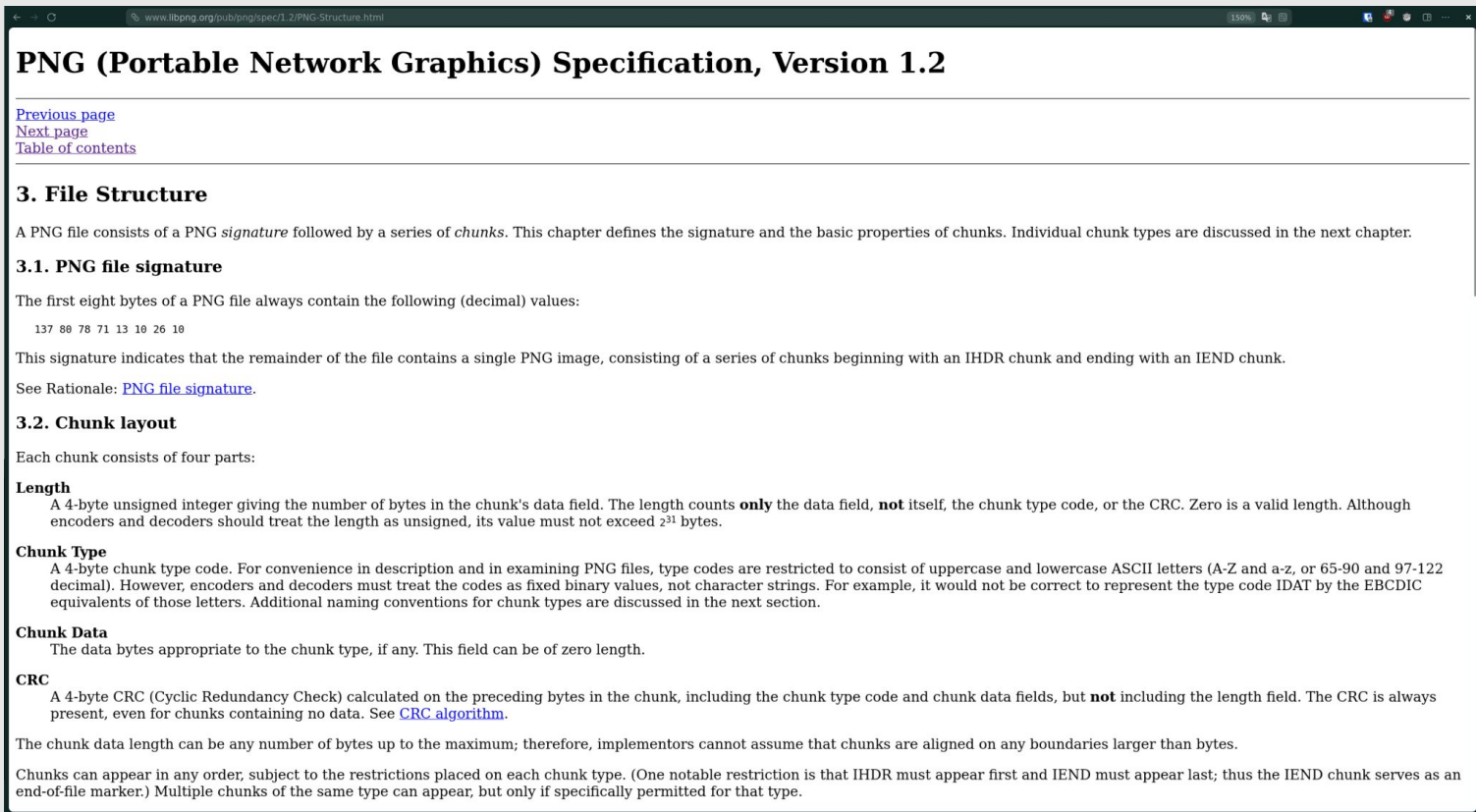
Just data... with a header

Encoding		Grammar	Cus
0	89 50 4E 47 0D 0A 1A 0A 00	.PNG .	Position Element Value
9	00 00 0D 49 48 44 52 00 00	.. IHDR..	0x00 PNG file [0]
2	00 0C 00 00 00 0C 08 06 00		0x00 Header 89 5
8	00 00 56 75 5C E7 00 00 00	..Vu\ç...	0x08 234 pad...
4	04 67 41 4D 41 00 00 D8 EB	.gAMA..øè	
D	F5 1C 14 AA 00 00 00 20 63	õ .ª... c	

And sometimes with a pretty structure

	0x0	0x1	0x2	0x3	0x4	0x5	0x6	0x7	0x8	0x9	0xa	0xb	0xc	0xd	0xe	0xf		
0x0000	Signature				Version		Flags		Compression		Mod:time		Mod:date		Crc-32			
0x0010	Crc-32		Compressed size			Uncompressed size			File name len			Extra field len						
0x0020							File name (variable size)											
0x0030							Extra field (variable size)											

What are files?

A screenshot of a web browser displaying the PNG Specification, Version 1.2. The browser's address bar shows the URL 'www.libpng.org/pub/png/spec/1.2/PNG-Structure.html'. The page title is 'PNG (Portable Network Graphics) Specification, Version 1.2'. Below the title, there are three links: 'Previous page', 'Next page', and 'Table of contents'. The main content area is titled '3. File Structure'. It explains that a PNG file consists of a PNG signature followed by a series of chunks. It then defines the signature and the basic properties of chunks. The section '3.1. PNG file signature' states that the first eight bytes of a PNG file always contain the following (decimal) values: 137 80 78 71 13 10 26 10. It also mentions that the remainder of the file contains a single PNG image, consisting of a series of chunks beginning with an IHDR chunk and ending with an IEND chunk. The section '3.2. Chunk layout' explains that each chunk consists of four parts: Length, Chunk Type, Chunk Data, and CRC. The 'Length' section describes it as a 4-byte unsigned integer giving the number of bytes in the chunk's data field. The 'Chunk Type' section describes it as a 4-byte chunk type code. The 'Chunk Data' section describes it as the data bytes appropriate to the chunk type. The 'CRC' section describes it as a 4-byte CRC (Cyclic Redundancy Check) calculated on the preceding bytes in the chunk. The document concludes by stating that the chunk data length can be any number of bytes up to the maximum, and that chunks can appear in any order, subject to the restrictions placed on each chunk type.

What are files? - Tools

Type based on the header

› file stt_logo.png

stt_logo.png: PNG image data, 1141 x 549,
8-bit/color RGBA, non-interlaced

Structure metadata

› exiftool stt_logo_sus.png

ExifTool Version Number : 13.58
File Name : stt_logo_sus.png
Directory : .
File Size : 23 kB
File Modification Date/Time : 2014:10:18 01:01:01+00:00
(...)
Comment : How would one even read this??

Find headers inside other files

› binwalk stt_logo_sus.png

DEC	HEX	DESCRIPTION
0	0x0	PNG image, total size: 11568 bytes
11568	0x2D30	PNG image, total size: 11567 bytes

Analyzed 1 file for 85 file signatures (187 magic patterns)

Strings

› strings imgs.zip

stt_logo.pngUT
zWTTK
CV :d
u{[z
SrwJ
(...)

What are files? - Tools

File specific

› webpinfo stt.webp

RIFF HEADER:

File size: 20576

Chunk VP8X at offset 12, length 18

ICCP: 0

Alpha: 1

EXIF: 0

XMP: 0

Animation: 0

Canvas size 600 x 420

Chunk ALPH at offset 30, length 1262

(...)

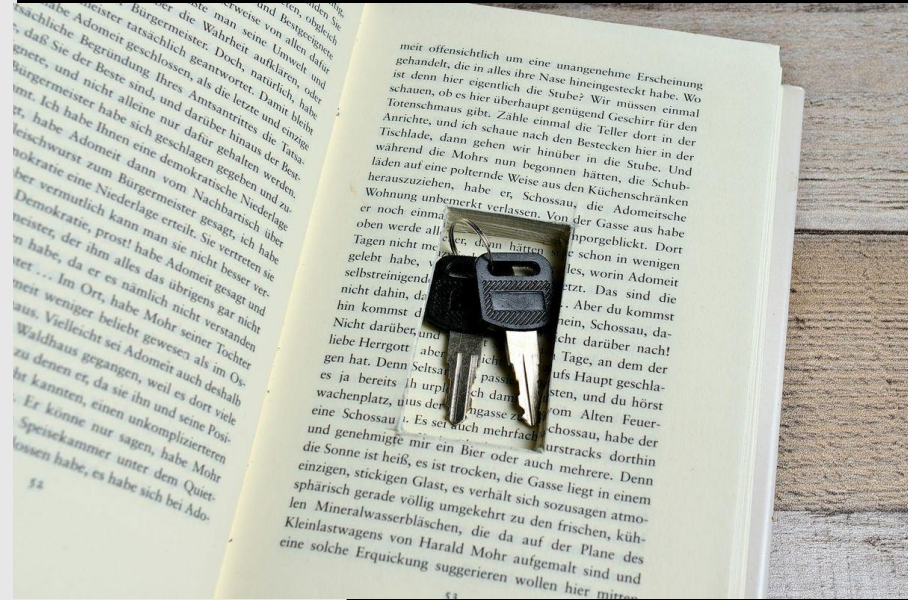
No error detected.

What's Stego?



Encryption

VS



Steganography



IMAGE STEGANOGRAPHY

What are images?

A matrix of pixels!

`Pixels[width,height] = img.load()`

And what are pixels?

Red, Green, Blue, Alpha (0-255)

`Pixels[x,y] = r,g,b,a`

`Pixels[x,y] = r,g,b`

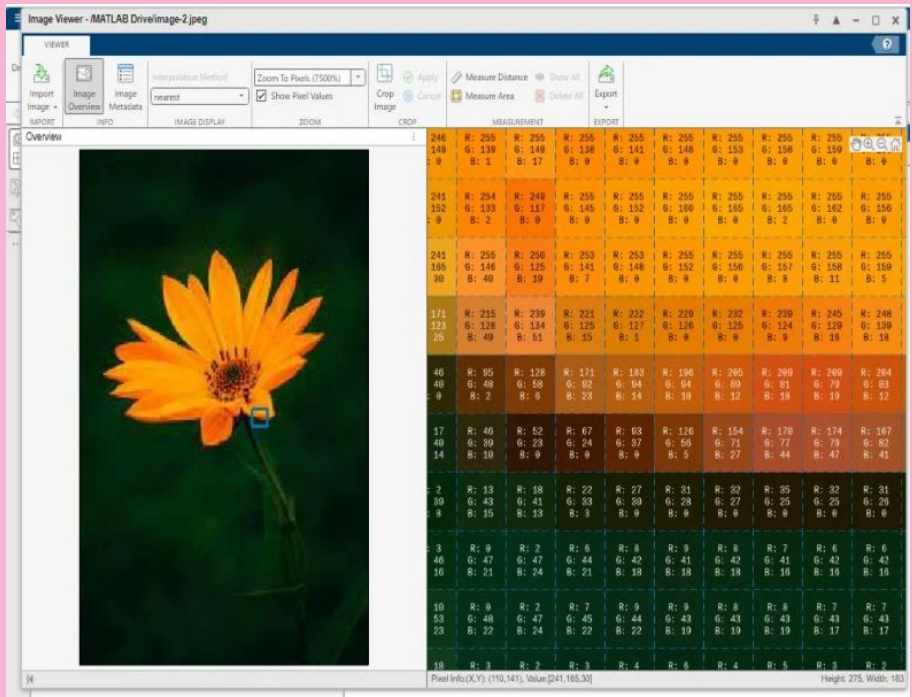
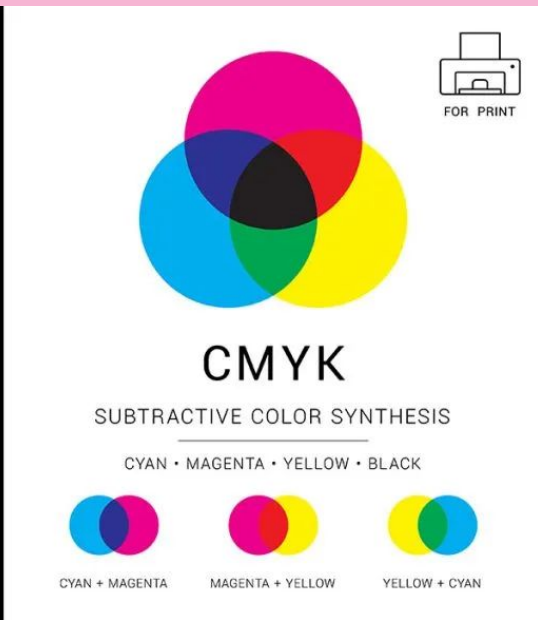
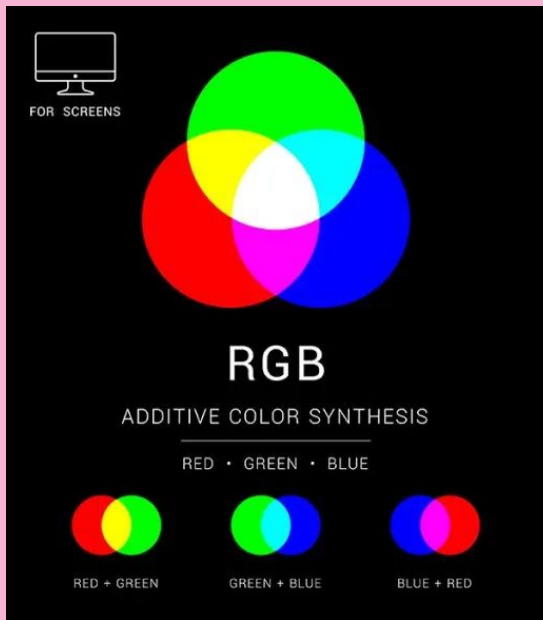


IMAGE STEGANOGRAPHY

RGB is just one color mode!



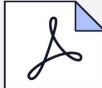
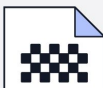
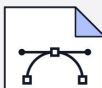
Best File Types for RGB	Best File Types for CMYK
 JPEG JPEGs are popular for digital files since they are a middle ground between file size and quality, and have good compressibility.	 PDF PDFs are ideal for print projects as they are compatible with most formats. They can be used to save bitmaps, vectors and combination files.
 SVG SVG files assign code to each shape in your design that can be read by your browser or app. They can be dynamically resized to fit any screen with no loss of quality.	 AI AI (Illustrator) is a vector design program, able to create images that can be resized with no loss of quality. These are great for large designs meant for poster or screen print.
 PNG PNGs are small-size file types that allow for image transparency. Consider these if you need to superimpose elements like buttons or graphics onto your web page or app.	 EPS EPS is a vector file type that's compatible with most vector programs. Use these when sending vector designs to a commercial printer.
 GIF GIFs are relatively small-size files that can display animated images in a browser or app. Animated logos or other small moving images will be best saved this way.	 TIFF TIFF (TIF) images are preferred for high-resolution photos and are often used by photographers who may blow up their images for print

IMAGE STEGANOGRAPHY

MSB & LSB

Tools: zsteg, stegsolve

1 LSB

Pixel[x,y] = R, G, B
= 01011011, 00111100, 10110101
newPixel[x,y] = R, G, B
= 01011010, 00111101, 10110101



4 LSB

Pixel[x,y] = R, G, B
= 01011011, 00111100, 10110101
newPixel[x,y] = R, G, B
= 01010010, 00111011, 10111110



1 MSB

Pixel[x,y] = R, G, B
= 11101010, 01100101, 10111011
newPixel[x,y] = R, G, B
= 01101010, 11100101, 10111011



U = U

IMAGE STEGANOGRAPHY

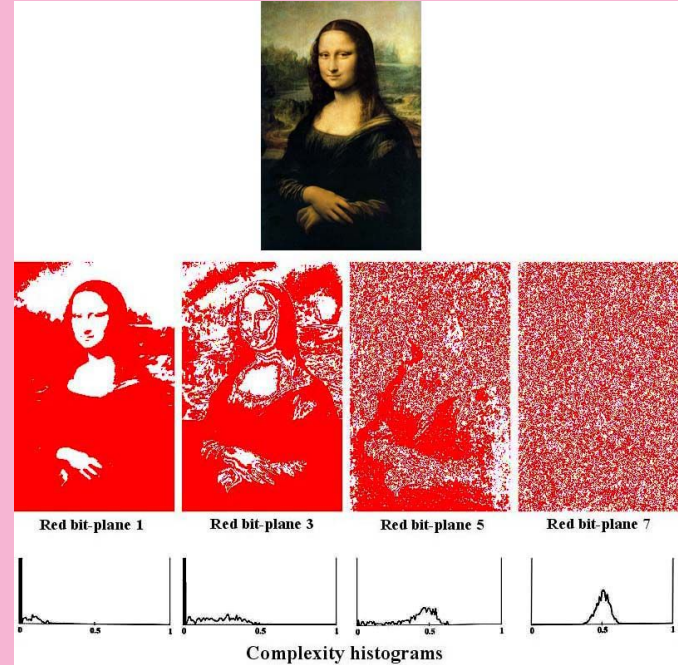
Bit planes

$R, G, B = \text{Pixel}[x,y]$

$\text{newPixel}[x,y] = R[7]$

(fill the pixel when Red's 7th bit == 1)

use StegSolve!



U = U

AUDIO STEGANOGRAPHY

What are audios?

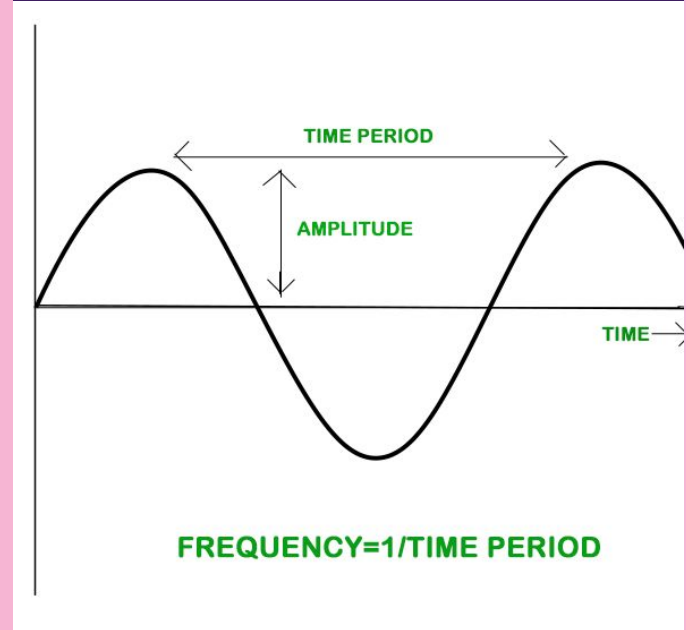
A sequence of amplitude samples!

$\text{samples}[\text{freq} * \text{time}] = \text{amplitude}$

$\text{Frequency} = \text{n_samples} / \text{second}$

A file with 44.1khz contains 44100 samples per second!

$\text{amp_after_2s} = \text{samples}[44100 * 2]$



AUDIO STEGANOGRAPHY

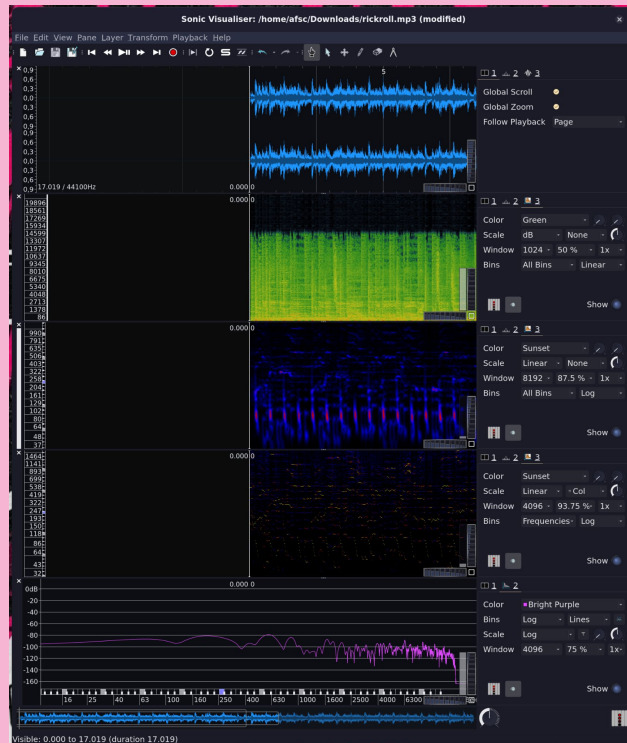
Data sequences can be visualised!

x =index, y =value

And much more ways...

Visualise: Sonic Visualiser

Edit: Audacity



A normal rickroll

Steganography

Sometimes it's all about a random tool....

